Project Name: FraudX – Real-Time Fraud Detection System

1. Introduction

As digital payments rapidly increase across platforms like UPI, credit/debit cards, and online banking, the risk of fraud has also grown significantly. Users are often targeted through phishing links, fake payment requests, and identity spoofing.

Currently, most systems detect fraud **after the transaction is completed**, making recovery difficult. This highlights the need for a **real-time fraud detection system** that can identify suspicious activities **before transaction completion**.

FraudX aims to address this issue by building an intelligent system that analyzes transaction behavior, detects anomalies, and prevents fraudulent transactions in real time.

2. Stakeholders

- **End Users (Customers):**
  Individuals performing digital transactions who need protection from fraud.
- **Banks & Financial Institutions:**
  Responsible for securing transactions and minimizing financial losses.
- **System Administrators:**
  Monitor fraud alerts and manage system operations.
- **Payment Platforms (UPI Apps):**
  Integrate fraud detection to enhance user trust and security.

3. Problems Identified

- **Delayed Fraud Detection:**
  Fraud is detected after transaction completion.
- **Lack of Behavioral Analysis:**
  Systems rely on static rules instead of user behavior patterns.
- **High Risk in Collect Requests:**
  Fraudsters exploit request-based payment systems.

- **Low User Awareness:**
  Users unknowingly approve fraudulent transactions.
- **No Real-Time Decision Making:**
  Lack of instant risk evaluation before transaction approval.

4. Existing System & Its Limitations

- **Rule-Based Detection:**
  Simple rules can be bypassed easily.
- **Post-Transaction Analysis:**
  No prevention before money transfer.
- **Limited Context Awareness:**
  Device, location, and behavior are not fully analyzed.
- **No Explainability:**
  Users are not informed why transactions are flagged.

5. Constraints & Assumptions

Constraints:

- Limited development time (hackathon)
- Use of mock/simulated data
- No real banking API access
- Limited computational resources

Assumptions:

- Users have consistent behavior patterns
- Fraud transactions deviate significantly
- System can simulate real-time processing
- Users respond to alerts and verification